

Topic 11: Real-World Applications of Generative AI

11.1 Workplace Productivity

- Writing assistance: drafting emails, reports, proposals, and meeting summaries
- Research and synthesis: summarising long documents, comparing multiple sources, answering questions about uploaded files
- Code assistance: GitHub Copilot, Cursor, and Codeium accelerate software development for developers of all skill levels
- Customer service: AI chatbots handle routine support queries around the clock and route complex issues to human agents
- Data analysis: natural language queries to databases and spreadsheets ('Show me revenue by region for Q3')
- Meeting tools: Otter.ai, Fireflies.ai, and Fathom automatically transcribe, summarise, and extract action items

11.2 Creative and Media Industries

- Graphic design: AI-generated concepts, logo variations, image editing, and social media visuals
- Music: AI composes background tracks (Suno, Udio), generates sound effects, and assists with mixing
- Film and TV: AI generates storyboards, concept art, visual effects, and dubbing in new languages
- Game development: AI generates textures, dialogue trees, NPC behaviours, and level design elements
- Publishing: AI assists with research, structural editing, cover design concepts, and marketing copy

11.3 Science and Research

- AlphaFold 2 (DeepMind, 2021): predicted the 3D structures of nearly all known proteins -- a 50-year problem solved; database publicly available at alphafold.ebi.ac.uk
- Drug discovery: AI identifies candidate molecules, predicts binding affinity, and flags potential toxicity -- reducing early-stage development time
- Materials science: AI models propose new materials with target properties such as improved battery performance or superconductivity
- Climate and weather: AI models (GraphCast, Pangu-Weather) produce weather forecasts faster and more accurately than traditional numerical methods
- Mathematics: AI assists in theorem proving; Google DeepMind's AlphaProof won silver-medal performance at the 2024 International Mathematical Olympiad

11.4 Healthcare

- Medical imaging: AI systems detect cancers, diabetic retinopathy, and fractures from X-rays and MRIs with performance comparable to specialist radiologists in specific benchmarks
 - Clinical notes: AI transcribes doctor-patient conversations and generates structured clinical notes automatically (e.g., Nuance DAX, Suki)
 - Drug discovery: Insilico Medicine used AI to identify a novel drug candidate for pulmonary fibrosis; it entered Phase 2 clinical trials -- a significantly compressed timeline
 - Genomics: AI analyses genetic variants to support personalised treatment decisions
 - Caution: all clinical AI tools require regulatory approval and human oversight; AI does not replace clinical judgement
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